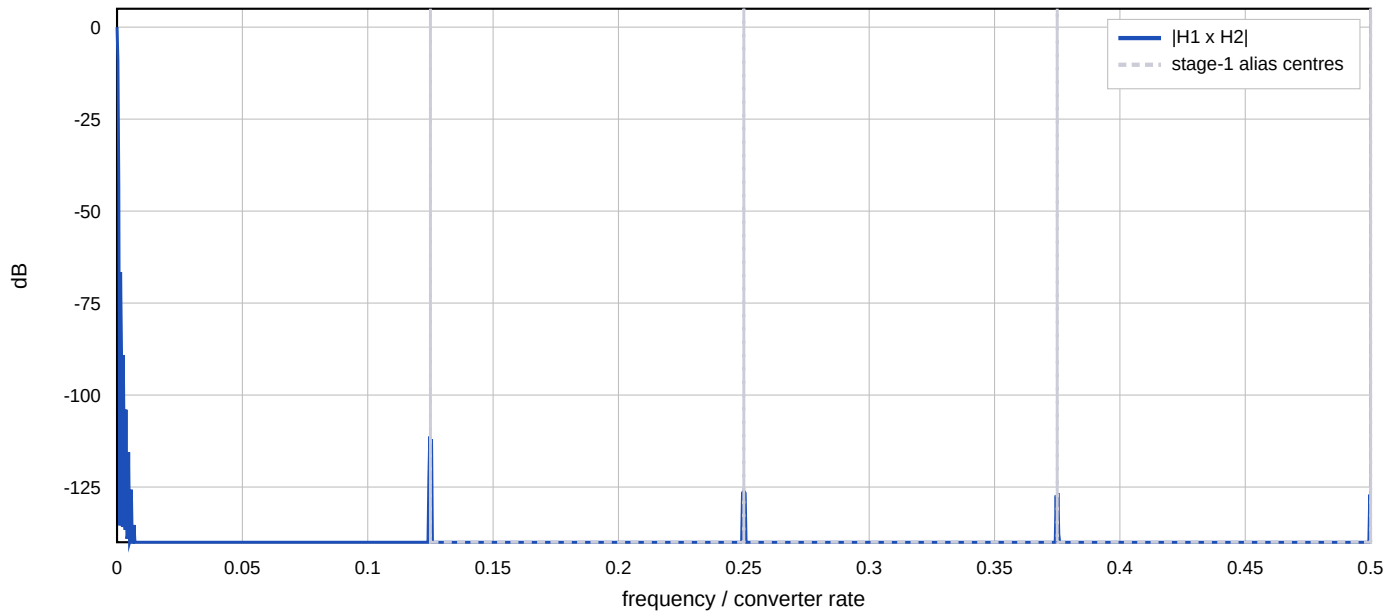


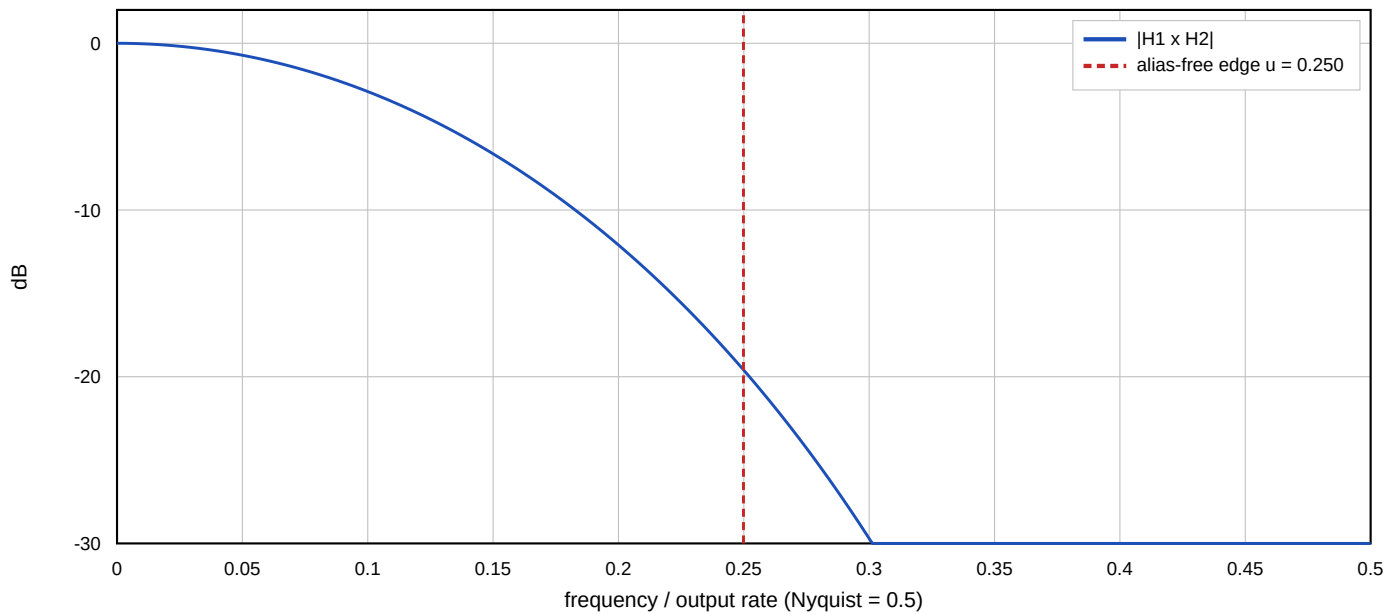
Decimation Chain - Derived Design

125.000 MHz / 488 = 256.1475 kHz, split [8, 61], 64.000 kHz alias-free

Combined magnitude, full input band



Combined magnitude across the decimated band



Stages

stage 1: /8 N=2 M=1 at 125.000 MHz

accumulator 22 bits, prune budget 12, widths [16, 16, 16, 16]

64 register bits, in 16 bits out 16 bits

stage 2: /61 N=5 M=2 at 15.625 MHz

accumulator 51 bits, prune budget 47, widths [37, 30, 24, 18, 16, 16, 16, 16, 16, 16]

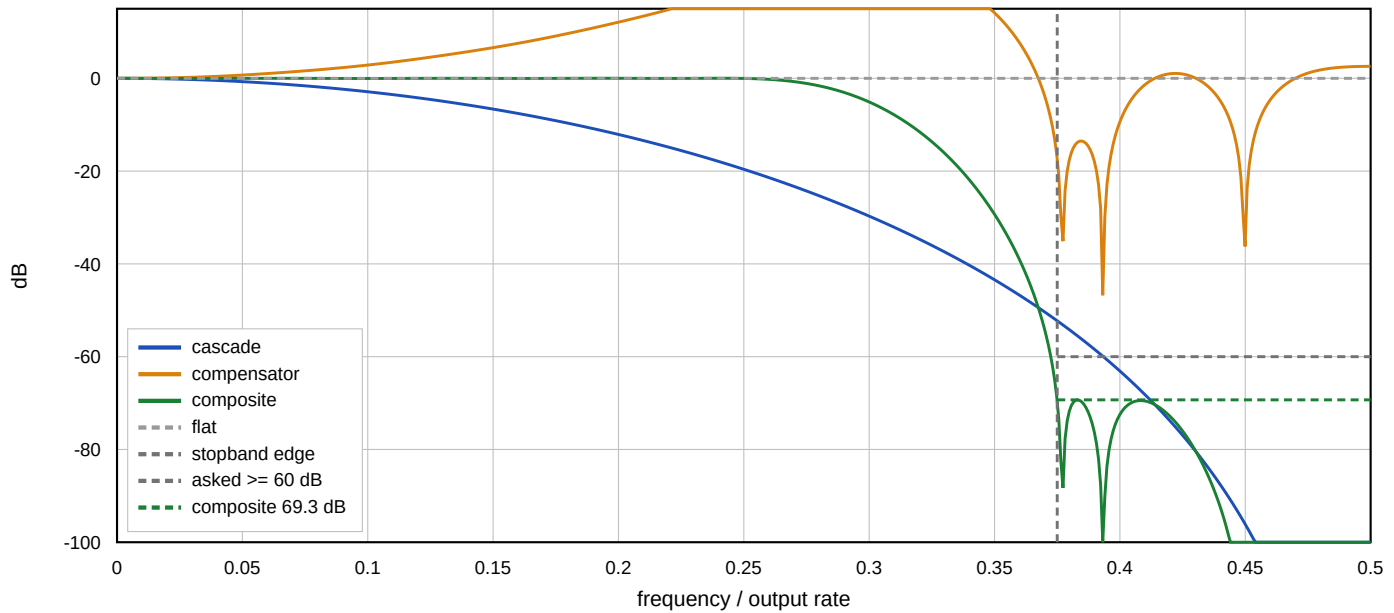
205 register bits, in 16 bits out 16 bits

A stage's nulls must reject the aliases of its own decimation: once energy has folded into the band no later stage can remove it, so every stage carries the full requirement.

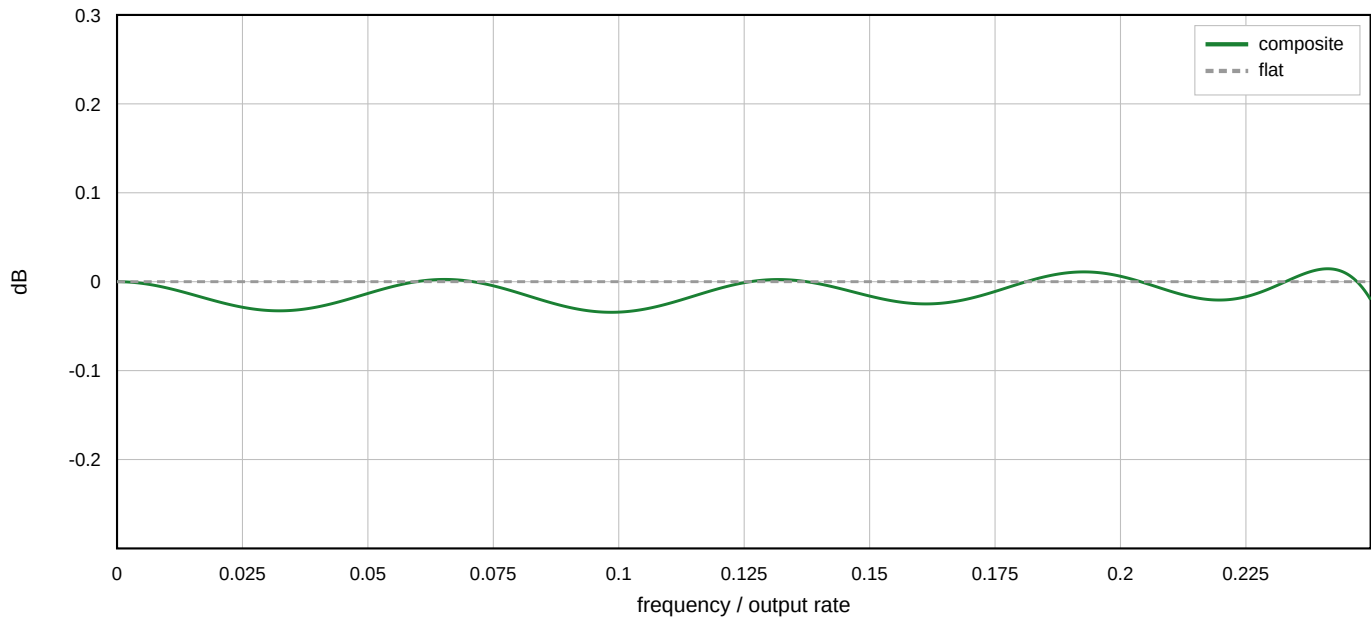
Compensation and Result

25 taps, 16-bit coefficients, 13 multipliers

Cascade, compensator, and the two together - full output band



Composite, zoomed - what is left



Achieved

ripple 0.0489 dB (asked <= 0.100)
alias rejection ... 67.3 dB (asked >= 60.0)
output SNR 84.6 dB (asked >= 80.0)
stopband 69.3 dB above 0.75 Nyquist (asked >= 60.0)

group delay 10278 input samples = 82.2 us
cascade 2427, integrator pipeline 33, comb pipeline 1960, output regs 2, compensator 5856
largest term is the compensator at 5856 (57%); loop bandwidth ~ 1.2 kHz

register bits 269 (rate-weighted cost 89.6)
taps [193, -717, 976, 180, -2545, 3236, 367, -5261, 3667, 6054, -11288, -1089, 16550, -1089, -11288, 6054, 3667, -5261, 367, 3236, -2545, 180, 976, -717, 193]
DC gain 1.000000 (exact by construction)
alternative [4, 61, 2], cost 96.8, 384 bits - feasible, but costlier by the rate-weighted model

Rate-weighted cost is a proxy, not an area figure: flops cost the same however slowly they are clocked. It captures that a wide adder at the converter rate is the hard part.